

Panorama Generation from Overlapping Photos: An Advanced Modular Pipeline

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Source Code & Dataset: <https://github.com/DanujaNirodhana2003/Panorama-generation>

Abstract—Panorama generation is a fundamental computer vision problem aiming to automatically stitch overlapping photos into a seamless panoramic image. Applications range from virtual tours and surveillance to autonomous driving and satellite imaging. This report presents the detailed design, mathematical foundations, implementation, and rigorous evaluation of an advanced, modular panorama stitching pipeline. The system was developed to construct wide field-of-view images from sequences captured using a handheld smartphone. We evaluate a baseline model utilizing ORB features against a finalized pipeline employing SIFT features, FLANN-based matching, and a smart-fallback RANSAC homography estimation. To achieve visual seamlessness, we implemented and compared three compositing techniques: simple overwriting, distance-transform feathering, and Laplacian pyramid multi-band blending. Finally, we integrated a global bundle adjustment module to mitigate cumulative alignment drift across sequential stitches. Our experimental results demonstrate that the finalized system achieves sub-pixel geometric accuracy (0.5468 pixels RMSE) and visually flawless blending, significantly outperforming the baseline methods. We present a thorough analysis of quantitative metrics, comparative blending visual results, mathematical models for geometric transformations, and an extensive discussion on remaining limitations.

Index Terms—Panorama, SIFT, RANSAC, Homography, Laplacian Pyramid Blending, Bundle Adjustment, Computer Vision, Feature Matching

I. INTRODUCTION

Panorama generation, also known as image stitching or image mosaicking, addresses the challenge of creating a larger field-of-view image from a sequence of smaller overlapping photographs. It is a widely studied and practically important technique in computer vision, with applications spanning virtual tourism, real estate visualization, autonomous driving, satellite and drone imagery analysis, and wide-area security surveillance [1].

The fundamental goal is to take multiple images of a scene—each captured from slightly different viewpoints—and combine them into a single, seamless composite. Achieving this requires solving several interrelated sub-problems: (1) detecting distinctive and repeatable visual features in each image, (2) establishing reliable correspondences between features in overlapping regions, (3) estimating the geometric transformation (typically a projective homography) that maps one image’s coordinate system onto another, and (4) warping and blending the images to produce a visually seamless result without noticeable seams or ghosting artifacts.

In this project, we developed a complete end-to-end panorama generation system. We began with a baseline implementation using the ORB feature detector and Brute-Force matching. While this baseline was computationally efficient, we identified severe limitations through systematic experimentation, notably high reprojection errors and visual artifacts. To overcome these, we iteratively improved the pipeline. The finalized pipeline incorporates SIFT features, approximate nearest neighbor matching, smart fallback transformation logic, advanced multi-band image blending, and global bundle adjustment.

This report is structured as follows: Section II discusses the related literature. Section III describes the dataset collection and preprocessing. Section IV details the modular software architecture. Section V provides an in-depth look at the mathematical and algorithmic methodology. Section VI presents the quantitative and qualitative experimental results. Section VII discusses failure cases, followed by the conclusion in Section VIII.

II. RELATED WORK

Feature-based image stitching has been extensively studied over the past two decades. Early approaches relied on corner detectors such as the Harris corner detector, which, while robust to rotation, lacked scale invariance.

The introduction of the Scale-Invariant Feature Transform (SIFT) by Lowe [2] revolutionized the field. SIFT provides keypoints that are invariant to image scale, rotation, and partially invariant to changes in illumination and viewpoint, producing robust 128-dimensional floating-point descriptors. While computationally heavy, it remains the gold standard for accuracy in structural matching. Faster binary descriptors like ORB (Oriented FAST and Rotated BRIEF) [3] were later introduced for real-time applications. ORB is significantly faster as it uses Hamming distance for matching, but it often sacrifices accuracy and scale invariance, making it less suitable for high-precision panorama stitching without strict constraints.

For matching high-dimensional descriptors like SIFT, exact nearest-neighbor searches are computationally prohibitive for large datasets. Muja and Lowe [4] proposed FLANN (Fast Library for Approximate Nearest Neighbors), which drastically reduces search time by automatically selecting between randomized KD-Trees and hierarchical K-Means trees.

To estimate geometric transformations robustly in the presence of outlier correspondences, the RANSAC (Random Sample Consensus) algorithm [5] is universally adopted. RANSAC iteratively samples minimal subsets of correspondences to compute candidate models, voting for the model with the highest consensus.

Image blending is the final critical step for creating a seamless panorama. Szeliski [1] discusses basic techniques like linear feathering, which smooths the transition between images based on the distance to the boundary. However, feathering can blur high-frequency details if the transition zone is too wide. To solve this, Burt and Adelson [6] introduced multi-resolution spline blending (Laplacian pyramids). This technique blends low-frequency colors over a wide area while keeping high-frequency structural edges sharp over a narrow area, producing superior seamless results.

When stitching many images sequentially, small geometric errors accumulate, causing a phenomenon known as cumulative drift. Triggs et al. [7] formalized Bundle Adjustment (BA), a global optimization technique that refines all camera poses simultaneously to minimize reprojection errors across the entire dataset.

III. DATASET & PREPROCESSING

A. Dataset Collection

A custom dataset was collected specifically for this project to ensure we had control over the capture parameters. Images were captured using a handheld smartphone camera with a 12-megapixel resolution. The dataset consists of multiple sequences of images (e.g., *test1* and *test2* sets) taken with a deliberate 30–50% horizontal overlap between consecutive frames.

The capture was performed outdoors on the university campus, featuring complex architectural structures, vegetation, and pathways. Images were captured at a fixed camera height with consistent lighting conditions to minimize exposure variation. However, natural outdoor lighting inevitably introduces some

differences, providing a realistic challenge for the blending algorithms.

B. Preprocessing Pipeline

The preprocessing pipeline is intentionally kept minimal to preserve original image data, consisting of the following steps:

- 1) **Grayscale Conversion:** Feature detectors (SIFT/ORB) operate on intensity gradients. Input images are internally converted from BGR to single-channel grayscale during the feature extraction phase, while the original color data is preserved in memory for the final warping and blending phases.
- 2) **Distortion Handling:** Given that modern smartphones have relatively low lens distortion and perform some internal digital correction, no explicit mathematical barrel or pincushion distortion correction was applied prior to processing.
- 3) **Resolution Management:** The final pipeline processes images at full native resolution. This takes advantage of SIFT’s scale-space extrema detection to find stable, high-quality features that might be lost if the images were aggressively downsampled.

IV. SOFTWARE ARCHITECTURE

To support the advanced features of the final submission and ensure maintainability, the codebase was entirely refactored from a monolithic script into a modular, production-ready architecture. The system is orchestrated by a central `pipeline.py` script, which dynamically invokes specialized modules:

- **`src/features/extractor.py`:** Handles the instantiation and execution of SIFT and ORB detection algorithms.
- **`src/matching/matcher.py`:** Implements FLANN matching logic, KD-Tree configurations, and Lowe’s ratio test filtering.
- **`src/geometry/homography.py`:** Manages homography estimation using RANSAC, calculates geometric stretching metrics, and executes the affine fallback logic.
- **`src/blending/`:** Contains distinct algorithms for `feathering.py` (distance transforms) and `multiband.py` (Laplacian pyramids).
- **`src/bundle_adjustment/optimizer.py`:** Provides global pose refinement matrices using least-squares optimization over the graph of image overlaps.
- **`src/stitching/warper.py`:** Computes canvas dimensions, bounding box translations, and executes the OpenCV perspective warping.
- **`src/evaluation/metrics.py`:** Automatically calculates RMSE, inlier ratios, and computational time for benchmarking.

This strict modularity allows for isolated unit testing (via the `tests/` directory) and parameter injection via YAML configuration files, ensuring the pipeline can be easily adapted for future research.

V. METHODOLOGY

A. Mathematical Foundations of SIFT

The finalized pipeline relies heavily on SIFT for feature extraction. SIFT identifies keypoints in a scale-space by convolving the image $I(x, y)$ with Gaussian filters $G(x, y, \sigma)$ at different scales. The Difference of Gaussian (DoG) function $D(x, y, \sigma)$ is computed between adjacent scales to find stable extrema:

$$D(x, y, \sigma) = (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \quad (1)$$

Local extrema in the DoG pyramid are selected as keypoints. A dominant orientation is assigned based on local image gradients, and a 128-dimensional descriptor vector is generated, making the feature invariant to scale and rotation.

B. Robust Feature Matching

Matching is performed using the FLANN matcher with KD-Trees (configured for 5 trees and 50 checks) to achieve logarithmic search complexity. To eliminate ambiguous matches, we apply Lowe's Ratio Test. For a given descriptor, let d_1 be the Euclidean distance to the nearest neighbor and d_2 be the distance to the second-nearest neighbor. The match is accepted only if:

$$\frac{d_1}{d_2} < \tau \quad (2)$$

where the threshold τ is strictly set to 0.70. This effectively purges matches that do not have a distinct, singular correspondence, vastly improving the quality of the data fed to the homography estimator.

C. Homography Estimation and RANSAC

The geometric relationship between two overlapping images is modeled by a 3×3 projective homography matrix H . It maps a point (x, y) in the second image to (x', y') in the first image:

$$c \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (3)$$

where c is a scaling constant.

Because feature matching is imperfect, we use RANSAC to estimate H . RANSAC iteratively selects 4 random point correspondences, computes a candidate H using the Direct Linear Transform (DLT) algorithm, and counts the number of inliers that satisfy a reprojection error threshold (set to 5.0 pixels). The number of iterations N required to ensure a high probability p of finding a model without outliers is given by:

$$N = \frac{\log(1 - p)}{\log(1 - (1 - e)^s)} \quad (4)$$

where e is the outlier ratio and $s = 4$ is the sample size.

D. Smart Fallback Transformation

A major limitation of standard homography estimation is unconstrained perspective distortion. When matching challenging images (e.g., low overlap), the resulting homography can cause the warped image to stretch uncontrollably toward infinity.

To solve this, we implemented a **Smart Fallback** mechanism. The system calculates the corner stretching factor of the proposed homography. If the bounding box width or height expands beyond a strict limit (e.g., $1.3 \times$ the original dimension), the pipeline discards the projective homography and falls back to a 2D Affine transformation. Affine transformations preserve parallelism and mathematically prevent infinite stretching, ensuring the pipeline never produces a completely distorted output.

E. Advanced Image Blending Algorithms

To achieve visual seamlessness, we implemented and evaluated three distinct blending modes:

- 1) **Overwrite (Baseline):** The warped source image is placed on the canvas, and the destination image is simply overwritten on top. While computationally trivial and perfectly sharp, it creates harsh exposure seams.
- 2) **Feathering:** We compute a Euclidean distance transform for both image masks. Let $D_1(x, y)$ and $D_2(x, y)$ be the distances from a pixel to the boundary of image 1 and 2, respectively. The blended pixel value I_{blend} is a linear combination:

$$I_{blend} = \frac{D_1 \cdot I_1 + D_2 \cdot I_2}{D_1 + D_2} \quad (5)$$

This smooths exposure differences but can slightly blur high-frequency details (e.g., leaves, window frames) in the overlap zone.

- 3) **Multi-Band Laplacian Pyramid:** The most advanced method [6]. The blending weights and both images are decomposed into a 5-level Gaussian and Laplacian pyramid. Low frequencies (base colors/lighting) are blended smoothly over a wide spatial area, while high frequencies (edges) are blended sharply over a narrow transition zone. The pyramid is then collapsed to reconstruct a perfectly sharp, seamlessly blended image.

F. Global Pose Refinement (Bundle Adjustment)

For sequential N-image stitching, pairwise homographies are multiplied iteratively ($H_{final} = H_1 \cdot H_2 \cdot H_3$). Small sub-pixel errors compound over time, leading to cumulative drift.

To combat this, we implemented a Bundle Adjustment (BA) module. After initially calculating all pairwise relationships, the BA module refines all H matrices simultaneously using the Levenberg-Marquardt algorithm. It minimizes the global reprojection error across all overlapping pairs:

$$\arg \min_H \sum_i \sum_j \| \mathbf{x}_{ij} - H_i \cdot \mathbf{x}'_{ij} \|^2 \quad (6)$$

This ensures structural consistency from end to end, even in panoramas spanning 5 or more images.

VI. EXPERIMENTS & RESULTS

A. Quantitative Evaluation

We rigorously evaluated the feature matching and geometric accuracy using the built-in `metrics.py` framework. Table I compares the deprecated ORB baseline with the finalized SIFT implementation.

TABLE I
QUANTITATIVE GEOMETRIC COMPARISON

Metric	ORB (Baseline)	SIFT (Final)	Improvement
Total Matches	37	384	10.4 $\times$ more
RANSAC Inliers	56.76%	76.56%	+19.8%
RMSE (pixels)	2.07	0.54	3.8 $\times$ better

The SIFT pipeline yields a 10.4 $\times$ increase in usable matches. Critically, the Root Mean Square Error (RMSE) drops from 2.07 pixels to 0.54 pixels. Achieving sub-pixel geometric accuracy is vital; it ensures that structural lines (like building edges) do not suffer from double-vision (ghosting) during the blending phase.

Figure 1 (spanning both columns) displays the high-quality feature correspondences generated by SIFT + FLANN after Lowe’s ratio test filtering. The parallel nature of the lines indicates a highly accurate geometric consensus.

B. Qualitative Blending Comparison

The most significant visual upgrade in the final pipeline is the blending architecture. We rendered the exact same scene using all three blending algorithms to demonstrate the progressive improvement and evaluate their efficacy in handling exposure differences.

As seen in Figures 2, 3, and 4, the progression from simple overwrite to multi-band blending resolves the visual discontinuities entirely. Multi-band blending successfully hides the stitch line in the clear sky while preserving the sharp structural integrity of the foreground foliage and pathways.

C. Multi-Image Stitching Results

Figure 5 demonstrates the capability of the pipeline to iteratively stitch multiple images into a wide field-of-view panorama. By leveraging global bundle adjustment to prevent drift and Laplacian pyramid blending to manage exposure variations across time, the system seamlessly integrates multiple sequential images.



Fig. 1. Feature correspondences between two overlapping high-resolution images. The clean, parallel match lines demonstrate the extreme effectiveness of SIFT feature extraction combined with FLANN matching and Lowe's Ratio Test in eliminating false positive outliers before RANSAC processing.



Fig. 2. **Overwrite (Baseline) Blending:** A harsh, visible seam line is clearly present down the center of the image. Because the smartphone camera automatically adjusted exposure between the two frames, placing one image directly on top of the other results in an unacceptable visual discontinuity.



Fig. 3. **Distance-Transform Feathering:** The harsh exposure seam is successfully smoothed out via linear distance-based weighting. The sky and ground blend naturally. However, upon close inspection, minor blurring (ghosting) is introduced in high-frequency regions (such as the leaves and branches) along the transition zone.



Fig. 4. **Laplacian Pyramid Multi-Band Blending:** The optimal and final result. Exposure differences in the sky and grass are seamlessly blended at low frequencies, while the structural edges of the trees and pathways remain perfectly sharp at high frequencies. No visible seams or blurring artifacts remain.



Fig. 5. **Multi-Image Stitching Pipeline in Action:** Sequential stitching of an extended panorama using distance-transform feathering and global bundle adjustment. The bundle adjustment ensures that the horizon remains level and geometric distortion does not compound exponentially across the image chain.

VII. FAILURE CASES & LIMITATIONS

Despite the advanced architecture, mathematical rigorosity, and high-quality results, certain fundamental limitations of image stitching remain challenging in specific scenarios:

- 1) **Moving Objects (Ghosting):** The current blending logic assumes a purely static scene. Subjects such as pedestrians or vehicles moving through the overlapping region during capture appear in different spatial locations across consecutive frames. The blending algorithms (even Laplacian pyramids) will attempt to blend these together, resulting in "ghosting" or duplicated, semi-transparent subjects in the final panorama. Advanced graph-cut seam routing is required to solve this by forcing the stitch line to route entirely around the moving object.
- 2) **Non-Planar Parallax Errors:** Homography mapping strictly assumes either a perfectly planar scene (e.g., a flat wall) or a camera rotating perfectly around its optical center (using a tripod with a nodal ninja). Because the smartphone was handheld, small translational movements occurred between frames. This causes parallax errors where close foreground objects and distant backgrounds move at different rates and cannot be simultaneously aligned by a single 3×3 matrix.
- 3) **Extreme Fallback Cropping:** While the Smart Fallback mechanism successfully prevents images from stretching to infinity due to bad homographies, falling back to an Affine transformation inherently limits perspective correction. If the scene actually required a strong perspective warp, the affine fallback will leave structural lines misaligned. Furthermore, the fallback can result in a smaller valid overlap area, forcing tighter rectangular crops in the final image to remove black borders.

VIII. CONCLUSION & FUTURE WORK

In this project, we successfully developed a robust, production-grade panorama generation pipeline. Transitioning from a simplistic ORB baseline to a sophisticated architecture utilizing SIFT, FLANN, Smart Fallbacks, and Bundle Adjustment allowed us to achieve sub-pixel alignment accuracy (0.54px RMSE).

Furthermore, by moving beyond simple pixel overwriting and integrating Laplacian pyramid multi-band blending, we completely eliminated visual seam lines caused by exposure variations, resulting in aesthetically flawless panoramas. The modular software architecture developed for this submission makes the system highly extensible for future research.

Future work would focus on implementing dynamic programming (graph-cuts) to find optimal stitch seams that deliberately route around moving objects, fully resolving the pedestrian ghosting limitation. Additionally, an automatic cylindrical or spherical projection step could be added prior to homography estimation to better handle 360° panoramas without severe edge stretching.

IX. INDIVIDUAL CONTRIBUTIONS

The project responsibilities were distributed equally among the group members, leveraging individual strengths to build the final modular pipeline. The specific contributions are detailed in Table II.

TABLE II
TEAM MEMBER CONTRIBUTIONS

Member	Contribution
K.D.N. Chandrasena (E/22/054)	Dataset collection, baseline ORB implementation, ROI-based feature extraction, spatial NMS filtering
K.P.B.P. Karunanayake (E/22/184)	SIFT+FLANN implementation, iterative stitching pipeline, modular refactoring, codebase architecture
K.J.V. Kahagalla (E/22/179)	Evaluation metrics implementation, bundle adjustment integration, quantitative analysis, final report writing
K.A.H. Kumarasinghe (E/22/205)	Homography estimation, smart fallback logic, multi-band blending implementation, presentation preparation

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